

How to use MADCUBA_STARS tool

Borja Barahona Gómez. Supervisors: Miguel Sanz Novo, Víctor M. Rivilla Rodríguez

May 12, 2026

This tool for MADCUBA (Martín et al., 2019) was developed to generate synthetic spectra of a subset of molecules stored in the file 'SAMPLESPECTRUM.spec'. It is based on a macro created in-house by Gabriel Jaimes (CAB) and identifies molecules in the template using MADCUBA macros, generating their corresponding simulated spectras as outputs. Further details about the tool are presented below.

1 Steps for running the tool.

1. Download the package of the tool. This package must contain a file .py called 'madcuba_stars.py', a textfile called 'madcuba_moleculardata.txt' and a reference spectra (i.e. 'SAMPLESPECTRUM.spec' in this example).
2. Change the path in which the files are located. In my case, I have put all my files in the Downloads directory. However, you can import all the files using other directories, only if you have defined them properly. For Linux, the syntaxis should be done by forward slashes (/), but in MACOS and Windows ¹ you should use backslashes (\). The places on which you need to change the code are: line 23, 24, 50, 53 and 56. In line 24 you need to specify a directory on which all the output data will be saved. This directory needs to have no documents, otherwise the tool will delete the directory.
3. Afterwards, open MADCUBA (download it [here](#)) and go to the horizontal table (or vertical, there is an option too) and click on *Macros* → *Run* and select the .py file you have downloaded. It will ask you (if you did not used it never) to download Jython (the language on which is based this script). You should download it and it will appear plenty of tabs on which you will find all the information you want: molecules detected, column density, catalog on which the data is defined...

2 Help for the tool.

- The script is made for modelling all the molecules contained in the textfile 'madcuba_moleculardata.txt'. You could modify this textfile adding or substracting molecules, always being consistent with the ordering and structure of the data.
- You can modify the temperature excitation, the FWHM (full width half maximum) or the local standard rest velocity v_{lsr} . In this example, we use $T_{ex.} = 10\text{ K}$ and $FWHM = 20\text{ km/s}$. You should not play with the variable *wantrefs*, as it should have only 2 possible values: 1 if you want the references of all the appearing molecules, and 0 if not.
- There is a system where the data is cleaned every moment you run other molecules. If you want to remove it, you have to put as a comment all the lines which have 'Delete'.

If you find any issues with the tool, please write to the mail bbarahon@ucm.es. Alternatively, you may reach out to the CAB supervisors at msanz@cab.inta-csic.es or vrivilla@cab.inta-csic.es.

References.

1. Martín, S., Martín-Pintado, J., Blanco-Sánchez, C., Rivilla, V. M., Rodríguez-Franco, A., & Rico-Villas, F. (2019). Spectral Line Identification and Modelling (SLIM) in the MADrid Data CUBe Analysis (MADCUBA) package: Interactive software for data cube analysis. *Astronomy & Astrophysics*, 631, A159. <https://doi.org/10.1051/0004-6361/201936144>

¹Using MADCUBA on Windows is not recommended because it has limited compatibility and less stable performance compared to Linux or MacOS environments.